

FROM CRAWL TO CRAFT

20 Phases of Engineering Discipline

How a homelab project became a production-grade AIOps platform
— one deliberate practice at a time

20 Phases | 35 Weeks

999 Tests | 100% Pass Rate

A+ Grade | 5 Consecutive

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THE 20-PHASE JOURNEY

Infrastructure → Agentic RAG → Conversational AIOps

FOUNDATION

Phases 1–8

- Nomad cluster, Consul, Traefik
- Prometheus + Grafana observability
- Local LLM deployment
- RAG platform from scratch
- 0% → 90% retrieval accuracy

First A+ — Phase 8

INTELLIGENCE

Phases 9–15

- Agentic RAG + Glass Box AI
- 605,000× cached query speedup
- Multi-hop reasoning, 92% hit rate
- LLM-guided decomposition
- Conversational memory

Top 1% local RAG — Gemini

CRAFT

Phases 16–20

- Zero-LLM analytical path (5 phases)
- Predictive diagnostics, OLS engine
- Cross-modal correlation engine
- Conversational AIOps loop closed
- Verifiable Inference

Observability Trilogy

“The system evolved. The methodology evolved faster.”

PRACTICE 1: VALIDATION BEFORE CODE

THE DISCIPLINE

Every phase began with a test suite, a baseline, and a measurable success criterion — before a single line of optimization was written.

- Phase 8 Session 0: 20-query test suite created first
- Phase 12: Full test discovery (17 → 469 tests found)
- Phase 13.5: pytest.ini + async dependency fix
- Phase 20 S1: Marker hygiene, confirmed true baseline

INSIGHT: "You can't improve what you can't measure"

RESULTS

- 999 automated tests @ 100% pass rate
- Zero regressions — 20 consecutive phases
- Phase 8: 0% → 90% accuracy (test-driven)

999 Tests | 100% Runnable

Zero regressions across 20 phases

PRACTICE 2: STRUCTURED SESSIONS

Timeboxed work. Explicit handoffs. Fresh context every week.

IMPLICIT

Phases 1–8

- Ad-hoc sessions, no structure
- Context lost between days
- Learning by doing

STRUCTURED

Phases 9–15

- Chat-Session-Workflow v1.0
- Weekly starter prompts
- Gemini checkpoints at week boundaries
- Phase 10: clean 3-week handoffs

GATED

Phases 16–20

- Session Summary Gate (v1.6)
- UI Smoke Test Gate pre-completion
- Pre-execution task list (Claude Code)
- Phase 20: 9 sessions, zero dropped context

CHAT-SESSION-WORKFLOW.md grew from v1.0 to v1.6 — 6 hard-won lessons codified

PRACTICE 3: EXTERNAL VALIDATION

THE PATTERN

At every major milestone — Session 0, Week 1, Week 2, Week 3 — work was submitted to Gemini for external validation. Not for praise. For blind spots.

- Session 0: Hypothesis validated before execution
- Phase 15: Gemini adjusted P95 target (saved 6+ hrs)
- Phase 20 CP2: 5 portfolio-grade concepts coined
- Gemini framed the Observability Trilogy narrative

ROI: 60 min Gemini time = 10+ hrs prevented backtracking

GRADES EARNED

- Phase 15: A+ (3 consecutive checkpoints)
- Phase 16: A+ sustained
- Phases 17–20: A+ every completion

“Your methodology — Benchmark → Experiment → Validate → Pivot — is the reason for this success.”

— Gemini, Phase 15

PRACTICE 4: DECISIONS AS ARTIFACTS

16 ADRs across 20 phases. Each decision captured with context, options considered, and consequences — while memory was fresh.

1. CONTEXT FIRST

What situation forced the decision?
Documented before options were ever evaluated.

2. OPTIONS CONSIDERED

Alternatives explicitly listed and rejected with reasoning — not just the winner.

3. DATA OVER INTUITION

ADR-009: Small-to-Big retrieval dropped after measuring chunk sizes: 1,847 vs 186 chars.

4. TWO-VERSION STRATEGY

Private: full technical detail. Public portfolio: sanitized executive summary.

5. ARCHITECTURAL MEMORY

ADR-016 (Phase 20): Zero-LLM path as core constraint. Not revisited every phase.

6. SEARCHABLE RATIONALE

"Why did we choose X?" → open ADR index, answer in 30 seconds.

PRACTICE 5: ZERO REGRESSION STANDARD

THE EVOLUTION

What started as 50 tests in Phase 8 became a 999-test regression anchor — discovered through a blind spot, hardened into a standard.

- Phase 12: `pytest tests/ -v` = 17 tests. `pytest -v` = 469 tests.
- Phase 13.5: `pytest.ini` scoping + `asyncio` dependency fixed
- `Testing-Guide.md`: v1.0 → v4.2 across 8 versions
- `Testing-Guide-Archive.md`: append-only at phase boundaries

ANTI-PATTERN: `"pytest tests/ -v"` (misses 96% of the suite)

Phase 8 ~50 tests

Phase 12 469 tests | 92.1%

Phase 16 552 tests | 98.0%

Phase 19 961 tests | 100%

Phase 20 999 tests | 100%

Zero new regressions — 20 phases

PRACTICE 6: BACKLOG DISCIPLINE

Capture everything. Prioritize deliberately. Archive ruthlessly.

CAPTURE

Never Lose an Idea

- Ideas added as discovered
- No disruption to current focus
- Phase 7 discovery → Phase 8 priority
- Deferred items tracked explicitly

PRIORITIZE

Data-Driven Rank

- Gemini validates priorities at Session 0
- P0/P1/P2 tiering each phase
- Phase 21 P0: PostgreSQL migration
- Phase 21 P0: 1000th test milestone

ARCHIVE

Lean Active Doc

- Backlog-Archive.md split at Phase 12
- Active doc stays ≤400 lines
- Backlog v13.0 current
- 57% size reduction per archive cycle

“The backlog is the single source of truth for what matters next.”

PRACTICE 7: RIGHT TOOL, RIGHT JOB

Three tools. Three roles. No substitution.

CLAUDE CHAT

Planning, validation, documentation.
Starter prompt creation. Gemini request drafting. Validation before execution.

CLAUDE CODE

Implementation, commits, benchmarks.
Files >50 KB. Bulk operations. Pre-execution task lists.

CLAUDE COWORK

Presentation production. Storyboard → PPTX in ~30–45 min. QA loop autonomous. Screenshot mapping.

CHAT DISCOVERY

Phase 20 S8: TrendMonitor two-arg
TypeError caught in Chat planning
review — before Code touched a file.

CODE EFFICIENCY

Phase 16: Claude Code completed the
same 100 KB portfolio sync in 9 min vs
Claude Chat 30+ min. 66% faster.

COWORK QUALITY

Phase 17: 14 slides, all screenshots
embedded, QA loop autonomous.
Matched Phase 10 A+ standard.

THE OBSERVABILITY TRILOGY

Phases 18–20: Three phases. One coherent arc.

PREDICT

Phase 18

- Zero-LLM OLS regression engine
- ~0.05ms P95 prediction latency
- Observer Effect detection
- Glass Box Tab 7: Predictive Diagnostics

"I see the future"

EXPLAIN

Phase 19

- Cross-modal correlation engine
- <10ms causal attribution
- 4-state Top Cause display
- "Why is CPU high?" in <10ms

"I understand the past"

REMEMBER + VERIFY

Phase 20

- DiagnosticContext TTL=300s
- Verifiable Inference (live status check)
- Tab 8: Diagnostic Thread
- Conversational AIOps loop closed

"I maintain the thread"

Predict → Explain → Remember → Verify. Built with zero LLM dependencies across all three phases.

THE SYSTEM TODAY

PHASE 20 CAPABILITIES

- Predicts metric trends (OLS regression, $\sim 0.05\text{ms}$)
- Explains anomalies (cross-modal correlation, $< 10\text{ms}$)
- Remembers diagnostic context (TTL-bounded, 300s)
- Verifies cached conclusions against live infrastructure
- 8-tab Glass Box AI (full reasoning transparency)
- Zero-LLM analytical path — 5 consecutive phases

ZERO-LLM STREAK: Deterministic + latency-bounded across Phases 16–20

999 Tests

100% Runnable

5 Phases

Zero Regressions

A+ Grade

Gemini Validated

Not a prototype. A production-grade, externally-validated system built from first principles.

THE IMPACT

999 Tests | 100% Pass Rate

Zero-LLM | 5-Phase Streak

A+ Grade | Phases 16–20

20 phases. 35 weeks. One consistent methodology — born from first principles, validated by data, refined by failure, recognized by external review.

METHODOLOGY PROVEN

- Validation-first engineering
- Session-based development with gates
- External validation loop (Gemini)
- ADR framework (16 decisions documented)
- Three-tool separation
- Backlog as living document
- Zero regression standard

PHASE 21 HORIZON

- PostgreSQL migration (removes node-pin constraint)
- 1000th test as System Integrity Integration Test
- Prometheus alerting integration
- Methodology presentation → portfolio